

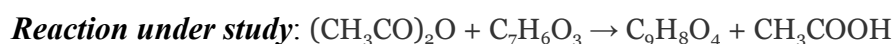
Introduction

Aspirin (acetylsalicylic acid) is an artificially synthesized compound derived from salicylic acid. Salicylic acid, naturally found in the bark of the willow tree, has been used since ancient times by the Greeks as painkillers (Eddy, 2003). Due to our bodies low tolerance to this compound, an alternative was researched. This led to the first synthesis of aspirin in 1897 by Felix Hoffman which became a household commodity ever since. In fact, the first synthesis and subsequent sale of aspirin is recognized for laying the foundation of modern pharmaceutical industries.

Today aspirin, like many other pharmaceutical drugs, are produced en-masse. As much as 40,000 tons of aspirin are synthesized each year (Fuster, Sweeny, 2011). Naturally, after getting over the initial shock of the amount of aspirin synthesized yearly, I became curious what conditions were used by factories for the synthesis of so much aspirin and whether or not they were optimal for the reaction. In chemistry when we learn about reactions we are merely presented with a “theoretical yield”. However, in reality reactions do not obtain such a yield as extraneous variables affect the process, giving a lower mass of product (called true yield). Some examples of these variables are: temperature, container of reaction, pressure, medium and physical state. Of these variables I was most interested in temperature's influence as it seemed to me to be the easiest to manipulate, compared to other variables, while simultaneously highly influencing the true yield of a reaction (Barry, 1914).

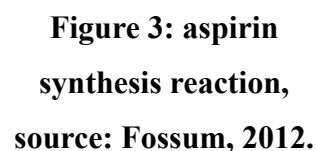
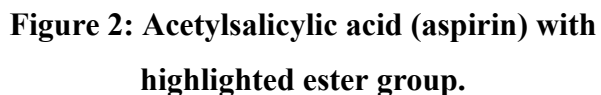
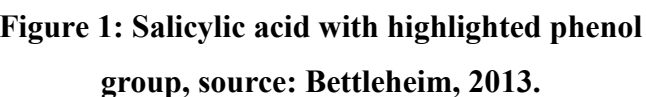
Temperature can be defined as a measure of the average kinetic energy of the atoms or molecules in a system. Basically, temperature shows us the average movement of molecules also known as vibrations, which influence the viability of a reaction taking place. Every reaction has its unique optimum temperature (mostly dependent on whether the reaction is exo- or endothermic), at which its reaction rate will be highest. This is desirable as a higher reaction rate allows us to synthesize more product in a constant timeframe. However, we cannot just increase the temperature of our reaction in an effort to increase its rate as, besides the additional cost, higher temperatures could begin to favor entirely different reactions than intended. This in turn, would ruin our product along with wasting our substrates.

There seems to be an intricate balance to be achieved between temperature, reaction rate and true yield which has led me to question what is the ideal temperature of the synthesis reaction of Aspirin, forming the research question: **How does the temperature of the synthesis reaction of Aspirin (60°C, 75°C, 90°C, 105°C, 120°C, 135°C) influence the yield and purity (measured by ferric chloride test) of the reaction?**



Background information

Salicylates are esters of salicylic acid. According to Drugs.com [10], a database used by institutions such as the Mayo Clinic, they have many useful properties, for instance they are antiseptic, antibacterial (and fungal), anti inflammatory and analgesic (pain relieving), to name a few. The phenol group in salicylic acid (highlighted below) is known to be irritating, which is why it is replaced with an ester group forming acetylsalicylic acid (aspirin).



**Figure 4: Hydrolysis reaction
of acetic anhydride, source:
Fossum, 2012**

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thus will create violet complexes. The intensity of the violet color will then be measured digitally and the concentration of molecules with phenol groups will be evaluated.

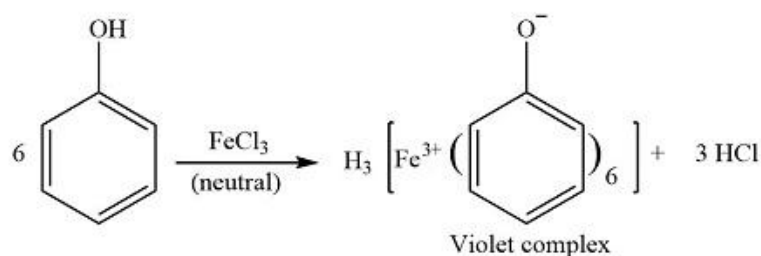


Figure 5: Phenol group converting into a violet complex by FeCl₃, source: [11]

Hypothesis

Higher temperatures increase the rate of a reaction along with its yield. However, Li and Zeng (2022) found that different temperatures affect the mechanisms occurring during high temperature reactions. This has led me to believe that when a certain temperature threshold is surpassed, other reactions besides the one intended may become viable, which would disrupt the intended chemical processes. Hence, I hypothesize that both yield and purity will primarily rise along with an increase in temperature as more substrates are allowed to react into products, however at higher temperatures, yield will drastically diminish while purity will remain indifferent thanks to the same purifying procedures taking place, which were able to successfully purify the product, while any foreign substances that were synthesized will likely not contain phenol groups so they will not impact the ferric chloride test results.

Variables

Preliminary study

The most common method for creating a stable temperature for reactions uses a water bath, whose effective temperature according to PennEHRS [12] reaches only around 70°C. Taking into consideration the sur-boiling temperatures of the experiment, this will not do as I need to achieve temperatures almost double that. According to PennEHRS: I could either use a silicone oil bath, salt mixture bath or a sand hot bath. I decided on using a sand hot bath as the high viscosity of silicone oil could prove to be a haphazard as any contact with extremities would not be easily removable (imagine a coating of oil on one's finger exceeding 100°C), the same going for salt mixture baths being in a liquid form, while any sand could easily be shaken off one's hand. During the pilot experiment I was able to successfully proceed with a 105°C trial using the sand bath. However, the fact that the bath was not a liquid proved to be a challenge as the temperature would vary based on the depth of thermometer placed in the sand. While on the other hand, I was not able to bury the flask deeper in the heated sand. Fortunately, the lab had a convenient electric chemical stove, whose temperature could be accurately set and stabilized. Furthermore, I would monitor the true temperature experienced by the reaction with a separate beaker placed next to (on the stove) the flask with the undergoing reaction.

Control variables

Variable	Control method	Justification
Stable temperature throughout the reaction	By using an electric chemical stove along with monitoring the temperature with additional thermometers.	In order to test the effect of specific temperatures on the reaction
Excess of acetic anhydride	By using a 72% excess of acetic anhydride, equal to 6 ml of 98% acetic anhydride.	In order to ensure that the limiting substrate is salicylic acid
FeCl ₃ concentration	By using the same 1% FeCl ₃ solution for all trials.	In order to ensure that the sensitivity of the FeCl ₃ complex is the same over trials
Salicylic acid mass	By carefully weighing the substrate and additionally calculating separate theoretical yields for each trial.	In order to ensure an equal reaction rate
Proper sterile procedures	By wearing sterile gloves, keeping a distance from the products and working under a fume hood.	In order to ensure the highest attainable level of purity.

Table 1. A list of variables being controlled in the experiment along with their method and reasoning.

Independent variable

Temperature of the synthesis reaction: 60°C, 75°C, 90°C, 105°C, 120°C, 135°C. An electric chemical stove will be used to heat up the reaction to the appropriate temperatures and a separate beaker with thermometers will be placed at the same time to control the true temperature.

Dependent variable

Yield of the synthesis reaction after recrystallizing the products weighed on a scale in grams along with the purity of the synthesized product evaluated by colorimetrically analyzing the intensity of the violet complexes created by FeCl₃ binding with phenol groups not present in pure aspirin.

Methodology

Apparatus and materials

Substances	Equipment		
Acetic anhydride 98%	Erlenmeyer flask (125ml), Lab book, Sharpie	Filter paper, Styrofoam box	Test tubes (25 ml), Test tube stand
Salicylic acid 100%	Graduated cylinder (± 1 ml)	Glass funnel, Aluminum foil	Paper towels, Pipettes (5ml, 50µl)
Sulfuric acid (18 mol · dm ⁻³), Iron(III) chloride	Science digital scale “Adam Equipment” (± 0.001 g)	Petri dish (150mm x 15mm), Timer (± 0.1 s)	Mini vortex mixer (VEVOR), Eppendorfs (1.5 ml)
Deionized water, Ice	Lab metal spoon, Glass baguette	Electric chemical stove “MS7” (± 1°C)	Fume hood (Enclosure)
Ethanol 95%,	Beaker (250ml),	Thermometer 0-250°C (± 0.5°C)	Plastic substrate trays (50 ml)

Table 2. A list of materials and equipment used throughout the experiment.

Safety, environmental and ethical considerations

Goggles and other safety equipment like a lab coat and gloves should be worn during the duration of the experiment. Acetic anhydride is irritating, especially to the nose. It should be under the hood at all times, making sure to avoid breathing in any vapors. Sulfuric acid used in the experiment can cause bad burns, handle it with caution. Excess chemicals should be disposed of in a tub of water. This will convert the acetic anhydride and dilute the sulfuric acid. Ferric chloride is both corrosive and toxic, do not handle it without gloves. Before utilizing any chemical substances, their safety data sheet should be first read and proper precautionary actions should be planned in case of an accident. Lastly, the aspirin synthesized will most likely not be pure enough to be taken internally. It should not be ingested.

Procedure

Begin by preparing all materials and apparatus for the experiment. Set up your work space under a fume hood and don your lab coat, gloves, safety glasses, and protective coat. Assemble the electric stove and heat up the metal tray. Numerically label the erlenmeyer flasks with a sharpie from 1-5 and note their temperature for the experiment in your lab notebook.

Start preparing the substrates by weighing 5g of salicylic acid using a science digital scale and transferring it to a 250mL Erlenmeyer flask using a plastic tray; remember to record the measured mass. Using a graduated cylinder, measure 7mL of acetic anhydride and transfer it to the flask using a serological pipette, exercising caution to avoid skin contact or eye exposure to vapors. Add 8 drops of $18 \text{ mol} \cdot \text{dm}^{-3}$ sulfuric acid as a catalyst to the flask. Swirl the flask delicately as to mix the substrates but not leave residue on the walls of the flask. Cover the flask with aluminum foil.

For heating the reaction, place the erlenmeyer flask onto the electric stove. Next to the flask, place a beaker along with two thermometers with a scale up to at least 200°C and monitor the temperature closely. Set a timer for 15 minutes and allow the reaction to take place.

After 15 minutes, turn the stove off and remove the flask. Allow the flask to cool under the fume hood for 3 minutes. Add 15mL of room temperature deionized water to facilitate excess acetic anhydride decomposition; swirl the flask. Cool the flask in an ice bath for 15 minutes to induce crystallization. If no crystals appear, scratch the inside of the flask with a glass baguette.

Moving on to isolating the product, set up a funnel in a beaker with filter paper, seat the paper with water, and collect the aspirin in the funnel by transferring the products from the erlenmeyer flask with a metal lab spoon. Rinse the flask with deionized water to remove as much product as possible and transfer the contents to the funnel. Allow the product to filter for 15 minutes.

For recrystallizing the aspirin, use a spoon to transfer as much aspirin from the filter paper as possible into a dry 250mL beaker. Label the beaker with a sharpie. Add 10mL of 95% ethanol to the beaker and warm the mixture to dissolve any crystals. Add 10mL of deionized water, cover the beaker with aluminum foil, and allow the solution to cool under the fume hood undisturbed for 10 minutes. Place the beaker into the ice bath until the mixture fully crystallizes.

Moving on to drying the purified aspirin, collect the aspirin in a petri dish from the beakers with a metal lab spoon and leave the petri dish open overnight under the fume hood to allow water and ethanol to evaporate.

Lastly, for analyzing aspirin quality, prepare a 1% ferric chloride solution. To do this, measure 0.1 g of ferric chloride using a science digital scale and transfer it to a test tube. Next, measure 9.9 ml of deionized water using a graduated cylinder and transfer it to the test tube with the ferric chloride. Using a mini vortex mixer, mix the contents of the solution until the ferric chloride dissolves. Then, put 2 aspirin crystals of each trial into a separate 1.5 ml Eppendorf. Add 1 ml of water using a pipette and a drop (10 ul) of 1% ferric chloride solution to the Eppendorf. Use a mini vortex mixer to properly dissolve the solution. Photograph the Eppendorfs on a flat, white tabletop, and analyze a 100 pixel by 100 pixel square in the center of the Eppendorf in <http://mkweb.bcgsc.ca/color-summarizer/>.

Raw Data

The theoretical yield of aspirin from the synthesis reaction was calculated using the following formula:

$$\text{Theoretical yield of aspirin} = \frac{\text{mass of salicylic acid}}{\text{molecular mass of salicylic acid}} \cdot \text{molecular mass of aspirin}$$

For instance the theoretical yield for the 60°C trial was calculated as so:

$$\text{Theoretical yield (60}^\circ\text{C)} = \frac{5.0011 \text{ g}}{138.13 \frac{\text{g}}{\text{mol}}} \cdot 180.17 \frac{\text{g}}{\text{mol}} = 6.5232$$

Temperature (± 1°C)	Salicylic acid mass (± 0.0001 g)	Combined weight of tube and aspirin (± 0.0001 g)	Weight of tube (± 0.0001 g)	Aspirin mass (± 0.0002 g)	Theoretical yield (± 0.0001 g)	True yield (± 0.03%)
60	5.0011	19.0821	14.2305	4.8516	6.5232	74.37%
75	4.9991	19.4605	13.9430	5.5175	6.5206	84.62%
90	4.9997	19.5468	14.2553	5.2915	6.5214	81.14%
105	5.0076	19.8592	14.2082	5.6510	6.5317	86.52%
120	5.0007	16.8489	14.0152	2.8337	6.5227	43.44%

Table 3. Raw Data obtained from the experiment with calculated yield.

The mean RGB value was calculated using the formula:

$$\text{Mean RGB value} = \sqrt{\frac{\sum(\text{squared single RGB value (ie. R) from all trials of a group)}}{\text{number of trials in group}}}$$

For instance the mean “R” value for the 60°C trial was calculated as so:

$$\text{Mean R value (60}^\circ\text{C)} = \sqrt{\frac{144^2 + 85^2 + 151^2 + 228^2 + 75^2}{5}} = 147$$

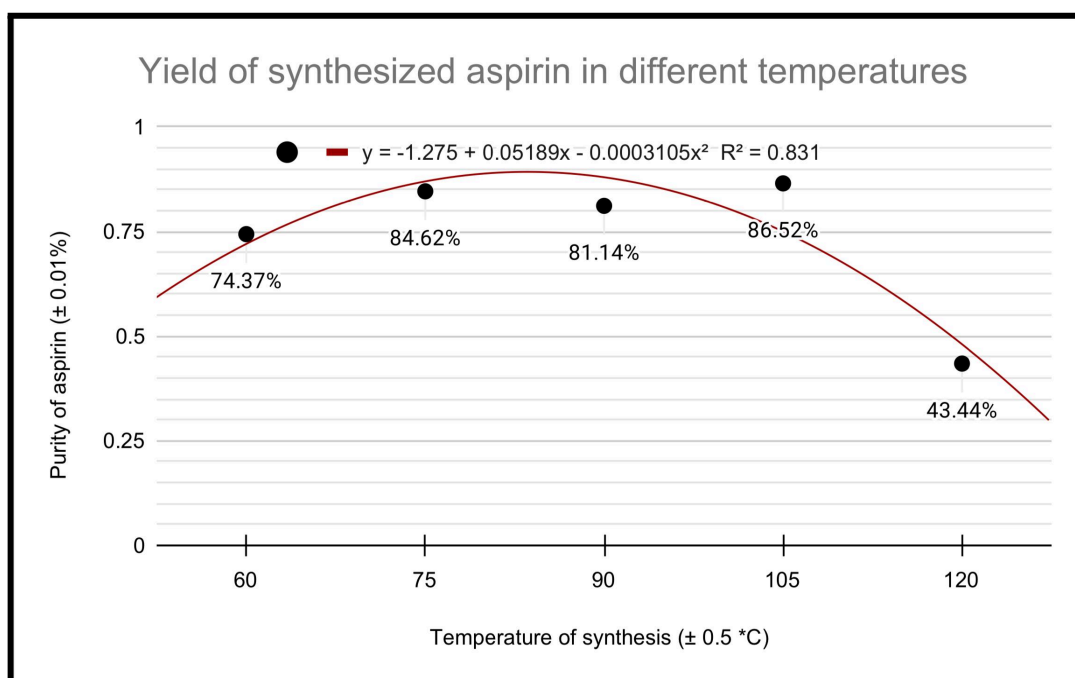
Temp. ($\pm$ 1°C)	RGB values					Mean	SD	Representation of mean
	Trial 1	Trial 2	Trial 3	Trial 4	Trial 5			
60	144 102 119	85 24 59	151 109 123	228 198 199	75 15 49	147 112 122	61 74 60	
75	121 82 101	74 23 49	66 19 46	57 10 39	62 13 44	79 40 60	26 30 26	
90	70 11 29	87 37 54	80 32 50	94 50 67	144 113 121	98 60 71	29 39 35	
105	144 112 126	92 48 71	62 11 40	159 122 136	63 12 41	112 77 92	45 53 46	
120	165 104 122	87 22 51	114 52 76	143 88 105	106 43 69	126 69 88	31 34 29	
Aspirin	84 54 74	65 34 56	65 35 56	61 27 52	90 63 81	74 45 65	13 15 13	
Salicylic acid	77 17 51	57 9 33	54 5 30	58 7 34	94 41 69	70 21 46	17 15 17	

Table 4. RGB raw data values obtained from color analysis of the samples with calculated mean and standard deviation (SD).

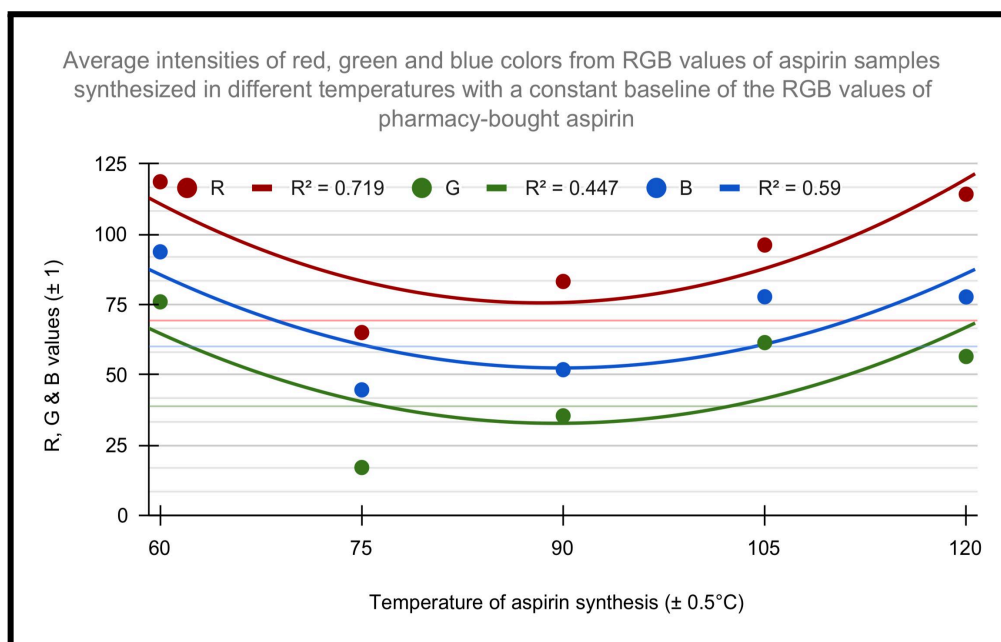
Qualitative observations

1. After the drying process, in all trials, the aspirin powder still seemed “puffy” as a result of all of the water not fully evaporating from the Petri dish.
2. After cooling the erlenmeyer flask from the 120°C condition, the contents were yellow, in contrast all other flasks had transparent contents after the heating process.
3. After filtering the 120°C condition the contents were clear like the rest of the conditions.
4. The dried aspirin powder from the 120°C condition had an odd, bubbly texture instead of the normal, snow-like texture found in other conditions.
5. Store-bought aspirin, used as a control, also came out positive (turned violet) in the ferric chloride test, although it was much more transparent compared to the salicylic acid control.

Processed Data



Graph 1: Graph representing the yield of synthesized aspirin in different temperatures with best-fit line, R^2 value and equation for quadratic regression line.



Graph 2: Graph representing the average intensities of red, green and blue colors from RGB values of aspirin samples synthesized in different temperatures with a constant baseline of the RGB values of pharmacy-bought aspirin, best-fit line and R^2 values for all three RGB colors, R: $y = 503.9 - 9.146x + 0.05048x^2$ G: $y = 404.6 - 7.47x + 0.03968x^2$ B: $y = 414.2 - 7.44x + 0.04x^2$.

While processing the data I noticed that pharmacy bought aspirin and salicylic acid had extremely similar RGB values (especially R and B). At first this seems counterintuitive as the ferric chloride test should show two entirely different results - one positive one negative, that's why I chose these samples for my baselines. However, when playing with the RGB values I observed that at a constant R and B value of around 70, 50 - playing with the G value in the range of 0-80 plays a significant role in how intense the violet shade is. Which makes sense, the red and blue intensity is constant, so they are the same hue, but shades induced by the level of green color.

Error propagation

Calculating percentage uncertainty of yield:

$$\% \text{ uncertainty of salicylic acid} = \frac{\text{uncertainty of apparatus}}{\text{value}} \times 100\% = \frac{0.0001}{5.6510} \times 100\% = 0.00177\%$$

Absolute uncertainty of yield = uncertainty of theoretical yield (uncertainty of salicylic acid) + uncertainty of aspirin measurement (uncertainty of test tube + uncertainty of tube with aspirin) = $0.0001\text{g} + 0.0002\text{g} = 0.0003\text{g}$

$$\% \text{ uncertainty of yield} = \frac{\text{absolute uncertainty of yield}}{\text{value}} = \frac{0.0003\text{g}}{0.8652\text{g}} \times 100\% = 0.0347\%$$

Calculating percentage uncertainty of purity:

$$\text{Absolute uncertainty of purity} = \text{uncertainty of RGB (sample)} = \{ 1, 1, 1 \}$$

$$\begin{aligned} \% \text{ uncertainty of purity} &= \frac{\text{absolute uncertainty of purity}}{\text{value}} = \left\{ \frac{1}{147}, \frac{1}{112}, \frac{1}{122} \right\} \times 100\% = \\ &= \{ 0.680\%, 0.893\%, 0.820\% \} \end{aligned}$$

Calculating the error of the experiment:

$$\text{Mean optimal purity temperature} = \frac{90.5^{\circ}\text{C} + 93.0^{\circ}\text{C} + 94.0^{\circ}\text{C}}{3} = 92.5^{\circ}\text{C}$$

Highest yield of aspirin synthesis was obtained at 83.5°C and purity at 92.5°C . To calculate the error of the experiment, results were compared with (Magbri, 2015) experiment, which showed that optimum temperature for synthesis is at 85°C .

$$\begin{aligned}\text{Yield value of error of experiment} &= \left| \frac{\text{Experimental value} - \text{Literature value}}{\text{Literature value}} \right| \times 100\% = \\ &= \left| \frac{83.5^{\circ}\text{C} - 85^{\circ}\text{C}}{85^{\circ}\text{C}} \right| \times 100\% = 1.76\%\end{aligned}$$

$$\begin{aligned}\text{Purity value of error of experiment} &= \left| \frac{\text{Experimental value} - \text{Literature value}}{\text{Literature value}} \right| \times 100\% = \\ &= \left| \frac{92.5^{\circ}\text{C} - 85^{\circ}\text{C}}{85^{\circ}\text{C}} \right| \times 100\% = 8.82\%\end{aligned}$$

Conclusion

The data obtained during the investigation of the synthesis reaction of Aspirin at various temperatures ranging from 60°C to 135°C , clearly show that temperature plays a vital role in influencing both yield and purity. The research question aimed to investigate how different temperatures affect the synthesis process of Aspirin, specifically focusing on yield and purity, measured through ferric chloride tests.

The hypothesis that yield and purity would rise and then fall was partially supported by the data. It seems that purity (measured by ferric chloride test) follows the trend of yield as can be seen on Graph 2. Graphs 1 and 2 showed that the optimum temperature for the yield of the synthesis reaction is 83.5°C ($\pm 0.5^{\circ}\text{C}$) while for purity: 92.5°C ($\pm 0.5^{\circ}\text{C}$). (The difference could be a sign of systematic error.) Graph 1 clearly shows a significant increase in yield as the temperature rises up to 83.5°C , followed by a decline thereafter. The data therefore isn't linearly correlated and a standard regression line can't be used. Instead, quadratic regression lines were used for both

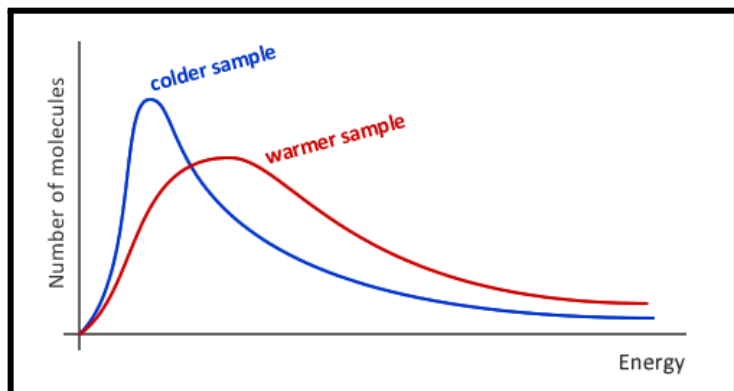


Figure 6: Two Maxwell-Boltzmann curves at different temperatures. The sample with the higher temperature has a higher amount of molecules with a higher temperature, meaning more molecules will surpass the activation energy of a reaction, source: Winfield, 2013.

Graphs 1 and 2. The initial rise in yield can be explained by the principles of the Maxwell-Boltzmann curve, which describes the distribution of speeds of particles and their energies at a given temperature.

Increasing the temperature affects the Maxwell-Boltzmann distribution by shifting the curve to the right, increasing the average kinetic energy of the molecules. This increase in kinetic energy leads to more frequent and energetic collisions between reactant molecules, consequently increasing the rate of the chemical reaction and subsequently, yield.

In the synthesis of Aspirin, as the temperature rises, molecules gain more kinetic energy, increasing collisions between the reactants: acetic anhydride, and salicylic acid. This increased collision frequency promotes the formation of the desired product, Aspirin, leading to higher yields (initially). However, after the 83.5°C threshold, the rate of reaction decreases due to the onset of decomposition reactions or the formation of undesirable byproducts.

The decomposition of Aspirin can occur at higher temperatures, especially above the optimum temperature range, leading to the breakdown of the desired product into byproducts such as acetic acid or phenol. Additionally, elevated temperatures can promote side reactions, where the reactants undergo alternative pathways resulting in the formation of impurities or unwanted compounds. These side reactions may include ester hydrolysis or acetylation of impurities present in the reaction mixture, forming for instance poly(salicylic acid) (PSA) (You, 2021), which would still contain the phenol at the end of the polymer chain and thus give a positive ferric chloride test result.

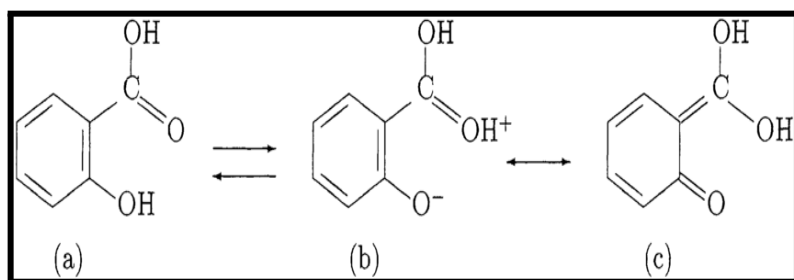


Figure 7: Salicylic acid (a) in zwitterionic form (b) as the resulting compound after a proton transfer reaction between the phenol group and the adjacent carboxyl group, which can be represented in the mesomeric structure related to the keto-tautomer of salicylic acid (c), which is an enol, source: Joshi, 2002.

Moreover, during the synthesis process, the formation of an enol intermediate is possible, particularly under specific conditions (Joshi, 2002). This highly reactive enol intermediate can further react with acetic anhydride to form a 1,3-diketone, potentially leading to the production of complex polyketides, which may exhibit coloration (Rat, 2021). This would explain the yellow contents found in the 120°C trial. The mere possibility of side reactions which lead to the formation of impurities highlight the importance of carefully controlling temperature and reaction conditions to maximize the yield and purity of the desired product while minimizing undesirable byproducts.

Research shows that 85°C is the optimal temperature for Aspirin synthesis, which compared to the findings of this investigation of an optimum of 83.5°C means the experimental error for yield was exceptionally low at less than 1.8%. This indicates high reliability of the results. However, the higher experimental error for purity at 9% raises concerns in regards to the method used for assessing purity, suggesting possible weaknesses in either the purification procedure or the antagonistic acetic anhydride presence test. The method should be fine-tuned to increase the accuracy of purity assessment.

Evaluation

The results of the experiment allowed me to answer the research question, and determine the relationship between the independent variable (temperature) and the dependent variables (yield and purity). The investigation deepened my understanding of the topic and its use in manufacturing processes. The experimental error for yield was found to be less than 1.8%, indicating the reliability of the results obtained. Furthermore, the optimal temperature is closely aligned with literature values, affirming the accuracy of the experiment. However, the experimental error for purity was approximately 9%, highlighting potential systematic errors of the methodology which should be extended.

Strengths:

The range of the independent variable tested was sufficient, with five distinct temperatures at even intervals, enabling the determination of relationships with the dependent variables. Additionally, conducting five repetitions while testing purity helped reduce random error and ensured more precise results. The use of a sand bath provided a stable environment for the reaction, allowing for higher temperatures than possible with water. Moreover, temperature monitoring with two thermometers ensured both accuracy and precision in temperature measurements.

Another strength lies in the approach to substrate measurement, which involved using a high mass of substrates with precise measurements. This approach helped keep the uncertainty percentage low, thereby reducing random error and increasing the reliability of the results obtained.

Weaknesses:

The result for optimum temperature for purity from the experiment (92.5°C) deviates from research by a substantial amount (85°C). Since the random error was low, repeating trials would not necessarily enhance the accuracy of purity measurements. Instead, to address the systematic errors a different method of color analysis, such as color spectrophotometry, could be used. Additionally, proper calibration of equipment and maintaining consistency in the experimental method would reduce error and maintain data accuracy.

Further accuracy in the experiment could be achieved by narrowing the data range now that we have confirmed the approximate optimal temperature for aspirin synthesis. Since literature suggested an optimal temperature of 85.0°C, the use of 15°C intervals prevented the in-depth investigation of yield and purity changes between 80°C and 90°C. Establishing measurements closer to the optimal temperature, with intervals every 2°C from 80°C to 90°C, would allow establishing discrepancies between experimental results and literature with higher accuracy. To further reduce random error during the experiment, increasing the number of measurements to 15 per trial would be beneficial. This would ensure that after detecting and excluding anomalous results we would still have over 10 measurements for calculating a precise mean and creating reliable graphs.

Another weakness identified in the method was the filtration process. The absence of suction during filtration resulted in longer filtration times and potentially left some solvent with the product, affecting the

determination of the final mass. Using a vacuum filter would address this issue by providing more efficient filtration and minimizing solvent retention, thereby improving the accuracy of yield measurements.

While purity testing through ferric chloride was replicated, the testing of crystals instead of a dissolved form in ethanol could have introduced inaccuracies. To enhance the reliability of the test, the crystals should be dissolved in ethanol at a 10% concentration before being mixed with the ferric chloride solution.

Furthermore, although pH was prepared uniformly, it wasn't controlled throughout the experiment. Given that pH can significantly impact the direction of the esterification reaction, it should be monitored using an electronic pH meter throughout the experiment. Addressing these weaknesses in the experimental methodology will contribute to the overall improvement of the experiment's reliability and accuracy.

Extending the method:

The experiment can be conducted under a constant temperature, instead manipulating the pressure of the reaction. While pressure typically doesn't affect liquid reactions due to liquids being considered incompressible, it can influence their boiling points. By adjusting the pressure, it may be possible to achieve higher reaction temperatures, which could enhance reaction rates and overall yield.

Another extension could be experimenting with different catalysts, primarily the difference between homogeneous and heterogeneous catalysts. They could reveal potential alternative reaction pathways or further optimize reaction conditions. Additionally, investigating the effects of different pH levels on the reaction could show how acidity/alkalinity influences the esterification process. Understanding the role of pH could help fine-tune reaction conditions for optimal aspirin synthesis.

Furthermore, examining potential byproducts of the aspirin synthesis reaction presents an opportunity to deepen our understanding of the process. Identifying and characterizing any byproducts generated during the reaction could shed light on side reactions or impurities that affect yield and purity. By gaining understanding of the formation and behavior of byproducts, researchers could create methods to minimize their formation and maximize the production of high-quality aspirin.

In summary, extending the experimental method to include manipulation of reaction pressure, testing different catalysts, varying pH levels, and investigating potential byproducts offers a pathway to refine and optimize aspirin synthesis. These extensions have the potential to enhance yield, purity, and overall efficiency of the synthesis process, contributing to advancements in pharmaceutical production and chemical synthesis techniques.

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